UNIT 3 · STANDARD 6.RP.1

Understand Ratios Flagship

Lesson 3-1-flagship

Name: ____________ Date: ____________ Class: ____________

CHEF ACADEMY MISSION

The Signature Recipe

You just earned your apron at Chef Academy. Head Chef Reyes is trusting you with her signature cookie recipe — 3 cups of flour for every 2 cups of sugar. Before service begins, you must read every recipe like a pro by mastering ratios, or the whole kitchen falls behind.

Key Vocabulary Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

3 cups flour to 2 cups sugar → 3:2

Ratio

A way to compare two amounts, like 3 to 2.

5 eggs vs. 3 eggs — which recipe uses more?

Comparison

Looking at two or more amounts to see how they are related.

3 cups flour : 2 cups sugar (ingredient to ingredient)

Part-to-part

A ratio comparing one part of a group to another part.

3 cups flour out of 5 cups total → 3:5

Part-to-whole

A ratio comparing one part to the whole group.

3 to 2 can be written as 3:2

Colon notation

Writing a ratio with two dots between the numbers, like 3:2.

Key Ideas & Notes

- Welcome to Chef Academy!
- Head Chef Reyes is teaching new students how to read recipes like a pro.
- Her signature cookie recipe calls for 3 cups of flour for every 2 cups of sugar.
- Before the students can start baking, they need to understand how these ingredient amounts compare to each other.
- Chef Reyes wrote several comparisons on the board. Sort them into two groups: statements that ARE ratios and statements that are NOT ratios.

Think About It

- What two ingredients are being compared in the recipe?
- How could you describe the relationship between flour and sugar?
- Is the amount of flour more than, less than, or equal to the amount of sugar?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes's signature cookie recipe uses 3 cups of flour for every 2 cups of sugar. How would you describe how these two ingredient amounts compare?

Level 1 support

Start here: The recipe uses 3 cups of flour for every 2 cups of sugar.

The recipe compares ____ to ____.

La receta compara ____ con ____.

For every ____ cups of flour there are ____ cups of sugar.

Por cada ____ tazas de harina hay ____ tazas de azúcar.

Word bank: Ratio | Comparison | Part-to-part

Listen for: *A strong answer names that flour and sugar are being compared and states the relationship as 3 to 2 (or 'for every 3 flour, 2 sugar'), not just that there is more flour.*

Level 2

Would the ratio still describe the same recipe if you wrote it as sugar to flour instead of flour to sugar? Justify why the order changes the ratio but not the recipe.

- The order matters because ____, so flour to sugar (____) is different from sugar to flour (____).
- Both describe the same recipe because ____.

EXPLORE

At the Prep Station you sorted the chef's statements. What made something a ratio instead of NOT a ratio (like '12 cookies on the tray')?

Level 1 support

Start here: A ratio compares two quantities, but '12 cookies' is only one amount.

This is a ratio because it compares ___ to ___.

Esto es una razón porque compara ___ con ___.

This is NOT a ratio because it only ___.

Esto NO es una razón porque solo ___.

Word bank: **Ratio** **Comparison** **Colon notation** **Part-to-whole**

Listen for: *Listen for the idea that a ratio must compare two quantities (using 'to', 'for every', 'out of', or a colon), while a single count or a description is not a comparison.*

Level 2

Critique this claim: 'The oven is set to 350°F is a ratio because it has the word to in it.' Is the student right? Defend your answer.

- The student is wrong because the word 'to' alone does not ___; a ratio must ___.
- 350°F is just ___, not a comparison of ___ and ___.

CONNECT

In Menu Planning the cafeteria wants 3 main dishes for every 2 side dishes. How is this the same kind of thinking as Chef Reyes's flour-to-sugar ratio?

Level 1 support

Start here: Both use a ratio to keep two amounts in the same comparison.

This is like our ratio work because ___ and ___ are compared.

Esto es como nuestro trabajo con razones porque se comparan ___ y ___.

The ratio of main dishes to side dishes is ___.

La razón de platos principales a guarniciones es ___.

Word bank: **Ratio** **Part-to-part** **Comparison** **Colon notation**

Listen for: *A strong answer identifies the 3:2 part-to-part ratio and explains that the cafeteria keeps that comparison the same even when amounts grow, just like the recipe.*

Level 2

If the cafeteria has 10 side dishes, generalize how you could figure out the main dishes — and explain why simply 'adding 1 more main than side' would NOT work.

- I can scale the ratio 3:2 by ___, so 10 sides means ___ mains.
- Adding 1 each time does not work because a ratio must ___, not ___.

Guided Examples

Example 1

A recipe uses 4 cups of milk and 1 cup of cream. What is the ratio of milk to cream?

Solution: A ratio compares quantities in the order given. Milk to cream = 4 to 1, or 4:1.

Answer: A. 4:1

Example 2

Chef Reyes uses 6 strawberries and 2 bananas in a smoothie. What is the part-to-whole ratio of bananas to total fruit?

Solution: Total fruit = 6 + 2 = 8. The part-to-whole ratio of bananas to total fruit is 2:8.

Answer: A. 2:8

Example 3

In a class of 30 students, 18 are girls. What is the part-to-part ratio of boys to girls?

Solution: Boys = 30 − 18 = 12. Part-to-part ratio of boys to girls = 12:18. (Order matters!)

Answer: A. 12:18

Write About the Math

The Writing Revolution

I can explain a comparison using the words ratio, part-to-part, part-to-whole, and colon notation.

1. Kernel Sentence

subject + verb

Model: Ratio is a way to compare two amounts, like 3 to 2.
Razón es una manera de comparar dos cantidades, como 3 a 2.

Write a kernel sentence about ratio. Use a subject and a verb.
Escribe una oración base sobre razón. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Ratio matters in math
Razón importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Ratio matters in math because ____.** *Razón importa en matemáticas porque ____.* _______________
but *pero*	**Ratio matters in math, but ____.** *Razón importa en matemáticas, pero ____.* _______________
so *entonces*	**Ratio matters in math, so ____.** *Razón importa en matemáticas, entonces ____.* _______________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about ratio.
Di un hecho verdadero sobre ratio.

Ratio ____.

Question
Pregunta

Ask a question about ratio.
Haz una pregunta sobre ratio.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about ratio.
Muestra entusiasmo sobre ratio.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with ratio.
Dile a un compañero qué hacer con ratio.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

The recipe compares ____ to ____.
La receta compara ____ con ____.

For every ____ cups of flour there are ____ cups of sugar.
Por cada ____ tazas de harina hay ____ tazas de azúcar.

This is a ratio because it compares ____ to ____.
Esto es una razón porque compara ____ con ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A parking lot has 10 cars and 6 trucks. What is the part-to-whole ratio of trucks to total vehicles?

A. 6:16

B. 10:6

C. 6:10

D. 16:6

Show your work:

2. A bag of marbles has 8 red, 6 blue, and 4 green marbles. Write three different ratios using these quantities: one part-to-part ratio, one part-to-whole ratio, and explain the difference.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Aliya's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A bag of trail mix contains 8 peanuts and 5 raisins. What is the ratio of raisins to total pieces?

A. 5:13

B. 8:5

C. 5:8

D. 13:5

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 6:16 *— Total vehicles = 10 + 6 = 16. The ratio of trucks to total is 6:16.*
2. **Try It 2:** Part-to-part: red to blue = 8:6. Part-to-whole: green to total = 4:18. The difference is that part-to-part compares two individual groups, while part-to-whole compares one group to the entire collection (8 + 6 + 4 = 18).
3. **Exit Ticket:** A. 5:13 *— Total pieces = 8 + 5 = 13. The ratio of raisins to total is 5:13.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Ratio is a way to compare two amounts, like 3 to 2.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).